

Erdős Problem #761

EP-761

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

The cochromatic number of G , denoted by $\zeta(G)$, is the minimum number of colours needed to colour the vertices of G such that each colour class induces either a complete graph or empty graph. The dichromatic number of G , denoted by $\delta(G)$, is the minimum number k of colours required such that, in any orientation of the edges of G , there is a k -colouring of the vertices of G such that there are no monochromatic oriented cycles.

Must a graph with large chromatic number have large dichromatic number? Must a graph with large cochromatic number contain a graph with large dichromatic number?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>